Nasser Mohammed

Department of Astronomy and Astrophysics
University of Toronto

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iD: 0009-0008-1224-0382

Research Interests: Stellar Streams, Dark Matter, Galactic Archeology, Astrostatistics, Data Science

Education

Sep. 2024 – Present **PhD in Astronomy & Astrophysics | University of Toronto**
Thesis topic: *Indirect Dark Matter Detection via Stellar Streams*
Supervisor: Professor Ting S. Li

Sep. 2019 – Dec. 2023 **BSc Honours in Physics | University of British Columbia, Okanagan**
Minor in Data Science
Thesis Title: *Polarimetric Study of 'Tadpole' Feature G137+7 with CHIME Radio Data*
Supervisor: Professor Alex S. Hill

Publications

First Author Publications

N. Mohammed, et al. 2026, submitted to ApJ: *The Kinematically Hot, Extremely Metal-Poor C-19 Stellar Stream in DESI DR2*

N. Mohammed, et al. 2024, ApJ, 971, 100: *Faraday Tomography with CHIME: The 'Tadpole' Feature G137+7*

Co-Authored Publications

E. Jarvis et al. (incl. **N. Mohammed**) 2026, submitted to ApJ: *Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon*

Awards & Honours

2026 NSERC Canadian Graduate Research Scholarship - Doctoral (CGSR D)
2025 NSERC Canadian Graduate Scholarship - Masters (CGS M)
2025 LSST Discovery Alliance Data Science Fellowship
2025 UofT Recognition of Excellence Award
2024 Mary Louise & Ronald Laidlaw Martin Graduate Scholarship in A&A
2024 David A. Dunlap Department of Astronomy & Astrophysics Entrance Fellowship
2024 NSERC Canadian Graduate Masters Scholarship (declined)
2024 NSERC Undergraduate Student Research Award
2023 Irving K. Barber Faculty of Science Undergraduate Research Award (2023)
2023 NSERC Undergraduate Student Research Award

Supervising & Mentoring Experience

Sep. 2025 – Present	Alex Brisebois	Graduate Mentorship Program | University of Toronto
Sep. 2025 – Apr. 2026	Maia Wertheim	Undergraduate Mentorship Program | University of Toronto
May 2025 – Sept. 2025	Alejandra Calderon	Research Project Co-Supervisor | Mitacs
Jan. 2025 – Oct. 2025	Isabelle Laing	Undergraduate Mentorship Program | University of Toronto
Apr. 2023 – Sept. 2023	5 Mentees	Globalink Mentorship Program | Mitacs

Mentored five international students travelling to UBC Okanagan for summer research projects.

Leadership & Service

Upcoming, Jul. 2026 — Local Organizing Committee Member: "Near Field Cosmology in the Era of Big Data: Local Group and Beyond" Workshop | The Fields Institute, University of Toronto

Oct. 2025 – Current — Union Steward | CUPE 3902 Unit 1
Union Steward for Unit 1 & Graduate Student Astronomy Association Union Representative.

Oct. 2024 – Current — Colloquium Speaker Dinner Organizer | David. A Dunlap Dept. of Astro. & Astrophysics
Co-ordinates post-doc and graduate student attendees, place reservations, accomodate speaker and attendee dietary restrictions.

Oct. 2024 – Oct. 2025 — Mediation Committee | Graduate Astronomy Student Association
Authored report on graduate student population financial wellbeing and work-life balance. Successfully advocated for increase in astronomy graduate student stipends.

May 2023 – May 2024 — Co-Founder & President of the UBC Okanagan Astronomy Club
Hosted seminars, coordinated stargazing nights, organized youth STEM outreach initiatives.

Further Education & Select Attendances

Feb. 2025 – Present — LSST Discovery Alliance Data Science Fellowship Program
Feb 2026 — RE:SciComm - Canada's Science Communication Conference
May 2023 — Canadian Astroparticle Physics Summer School, Queen's University & SNOLAB

Media Appearances

June 2025 — *Telescope images give 1st ultra HD images of the universe* | CBC News
On-camera interview; clip included in aired segment

Observing Experience

26 Sep. 2025 – 26 Sep. 2025 — Co-Investigator, Las Campanas Observatory | Región de Coquimbo, Chile
2 nights on Magellan/MIKE; Observations of the C-19 stellar stream

List of Talks & Lectures

May 2026 — AST101 Substitute Lecturer | University of Toronto
2 hr Lecture

Oct. 2025 — IAU Symposium 403: Hidden Beauty of the Galactic Outskirts | Córdoba, Spain
Oral presentation

Oct. 2025 — AST101 Substitute Lecturer | University of Toronto
1 hr Lecture

Aug. 2025 — York University EXPLORE Summer School | North York, ON, Canada
Lecturer

Jun. 2025 — Clusters in Halifax Workshop | Halifax, NS, Canada
15-minute talk

Jun. 2025 — CASCA Annual General Meeting 2025 | Halifax, NS, Canada
Poster presentation

Mar. 2024 — Okanagan College | Kelowna, BC, Canada
Guest lecture: Introduction to the Interstellar Medium (ASTR 121)

Feb. 2024 — University of Calgary PHAS Symposium | Calgary, AB, Canada
15-minute talk

Sep. 2023 UBC Okanagan Faculty of Science URA Symposium | Kelowna, BC, Canada
Poster presentation

Jun. 2023 CASCA Annual General Meeting 2023 | Penticton, BC, Canada
Poster presentation

Jun. 2023 TomFest: Frontiers in Radio Science and Engineering | Kaleden, BC, Canada
15-minute talk